Fatigue-Aware RIR-Based Load Prescription Model

This document summarizes the conclusions and model design developed for a fatigue-aware, RIR-based estimated 1RM (e1RM) and load prescription system suitable for use in a training application.

1. Core Principles

- Estimated 1RM must be anchored to near-failure sets (RIR ≤ 1).
- RIR should not be treated as extra reps inside classical 1RM formulas.
- Fatigue and capacity must be modeled separately.

2. Layered Model Architecture

- Base Capacity: A per-exercise base e1RM representing fresh strength.
- Fatigue Response: A per-exercise learned fatigue rate.
- Session State: A transient session fatigue value that resets each workout.

3. Base e1RM Calculation

- e1RM is updated only from hard sets (RIR ≤ 1).
- Brzycki formula recommended for rep ranges ≥ 6.
- Example: e1RM = weight / (1.0278 – 0.0278 × reps).

4. Intensity Factor (Reps + RIR → %1RM)

- Intensity factor = 1 – 0.025 × reps – 0.02 × RIR.
- This factor is used only for prescribing load, not estimating capacity.

5. Fatigue Modeling

- Session fatigue accumulates with each set.
- Set stress = (reps / 10) × (1 + (1 – RIR)).
- session_fatigue += fatigue_rate × set_stress.

6. Fatigue Discount

- fatigue_discount = exp(–session_fatigue).
- Effective e1RM = base e1RM × fatigue_discount.

7. Rest Timer Integration (Optional)

- If rest time is known, fatigue decays exponentially.
- session_fatigue *= exp(–rest_seconds / recovery_constant).
- If rest is missing, this step is skipped.

8. Freshness Bonus

- First set may receive a small probe bonus (≈3–6%).
- Used to test capacity early in session.

9. Load Prescription Formula

- target_weight = effective_e1RM × intensity_factor.

10. Learning & Adaptation

- Fatigue rate is updated based on performance error.
- fatigue_rate is bounded to avoid runaway behavior.
- Each exercise develops its own fatigue profile.

11. Design Guidance

- Fatigue variables should remain internal (latent).
- Users interact only with reps, RIR, and optional rest.
- Model functions with incomplete data.